

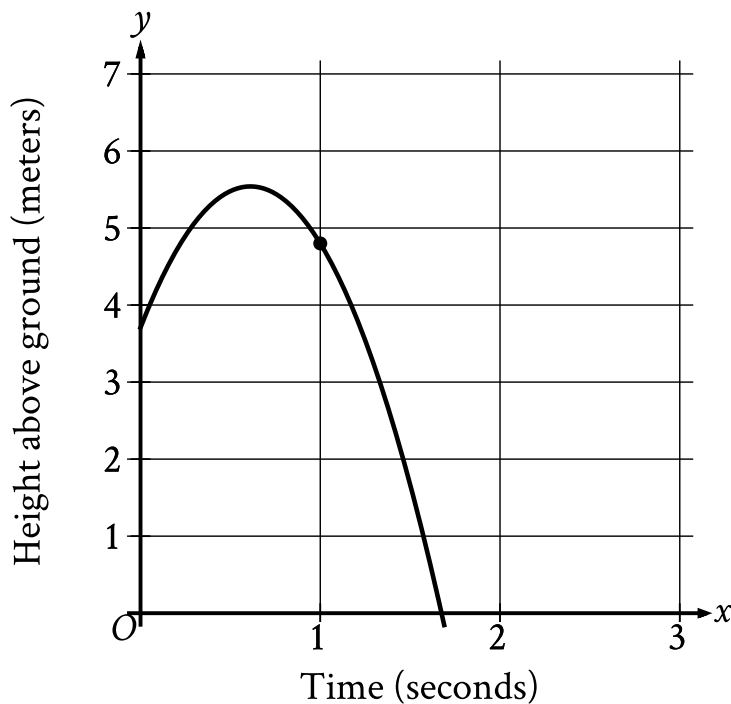
● EASY

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

03

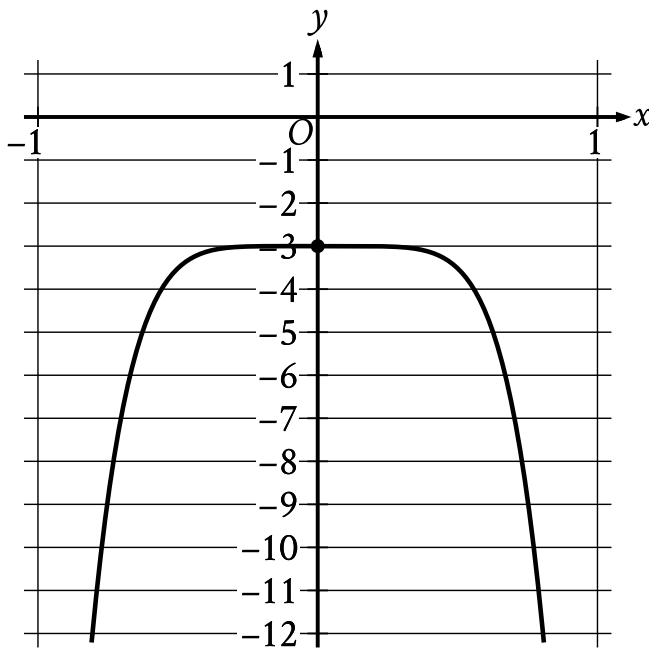


- All values are approximate.
- The parabola opens downward.
- The vertex of the parabola is at (0.6 comma 5.5).
- The parabola passes through the following points:
 - (0.0 comma 3.8)
 - (0.6 comma 5.5)
 - (1.0 comma 4.8)
 - (1.7 comma 0.0)

The graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. Which statement is the best interpretation of the marked point 1.0, 4.8 in this context?

- A. 1.0 second after being launched, the ball's height above ground is 4.8 meters.
- B. 4.8 seconds after being launched, the ball's height above ground is 1.0 meter.

- C. The ball was launched from an initial height of 1.0 meter with an initial velocity of 4.8 meters per second.
- D. The ball was launched from an initial height of 4.8 meters with an initial velocity of 1.0 meter per second.



- In quadrant 3, the curve rises to its maximum at point (0 comma negative 3).
- In quadrant 4, the curve then falls.

The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is 0, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6

$\frac{7}{8}$	$\frac{7}{8}$	$\frac{7}{8}$	$\frac{7}{8}$
$\frac{8}{9}$	$\frac{8}{9}$	$\frac{8}{9}$	$\frac{8}{9}$
$\frac{9}{10}$	$\frac{9}{10}$	$\frac{9}{10}$	$\frac{9}{10}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

If $f(x) = \frac{x^2 - 6x + 3}{x - 1}$, what is $f(-1)$?

- A. -5
- B. -2
- C. 2
- D. 5

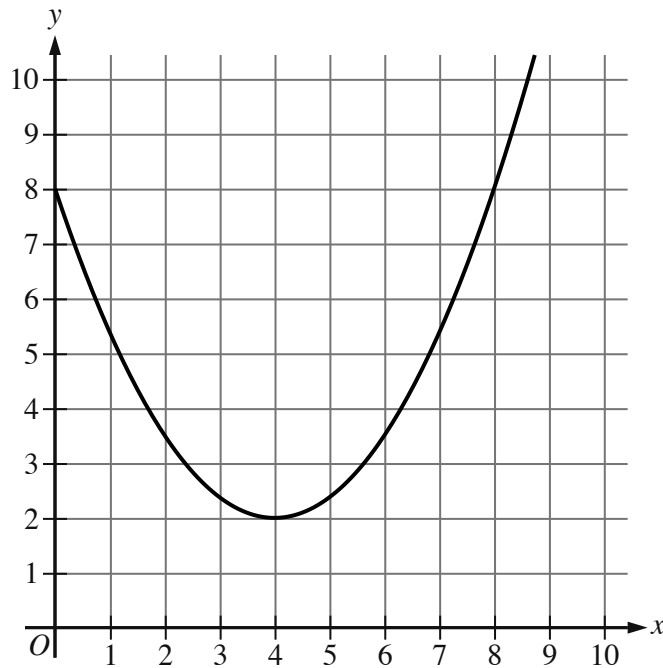
An investment account was opened with an initial value of \$ 890. The value of the account doubled every 10 years. Which equation represents the value of the account Mt , in dollars, t years after the account was opened?

A. $Mt = 890 \frac{1}{2}^{\frac{t}{10}}$

B. $Mt = 890 \frac{1}{10}^{\frac{t}{2}}$

C. $Mt = 890 2^{\frac{t}{10}}$

D. $Mt = 890 10^{\frac{t}{2}}$

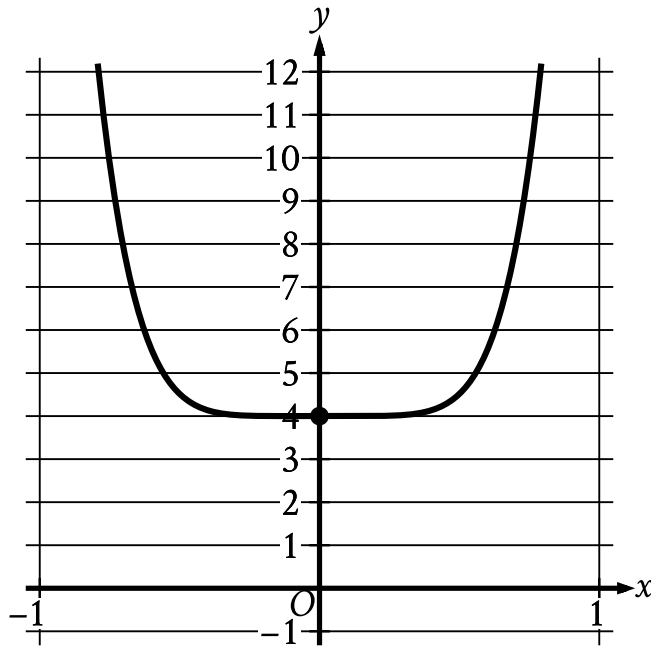


- The parabola opens upward.
- The vertex is at the point (4 comma 2).
- The parabola passes through the following points:
 - (0 comma 8)
 - (4 comma 2)
 - (8 comma 8)

The graph shows a marble's height above the ground y , in inches, x seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the y -intercept of the graph?

- A. The marble's height was 0 inches above the ground 8 seconds after it started moving.
- B. The marble's height was 8 inches above the ground when it started moving.
- C. The marble's minimum height was 0 inches above the ground.

D. The marble's minimum height was 8 inches above the ground.



- In quadrant 2, the curve falls to its minimum at the point (0 comma 4).
- In quadrant 1, the curve then rises.

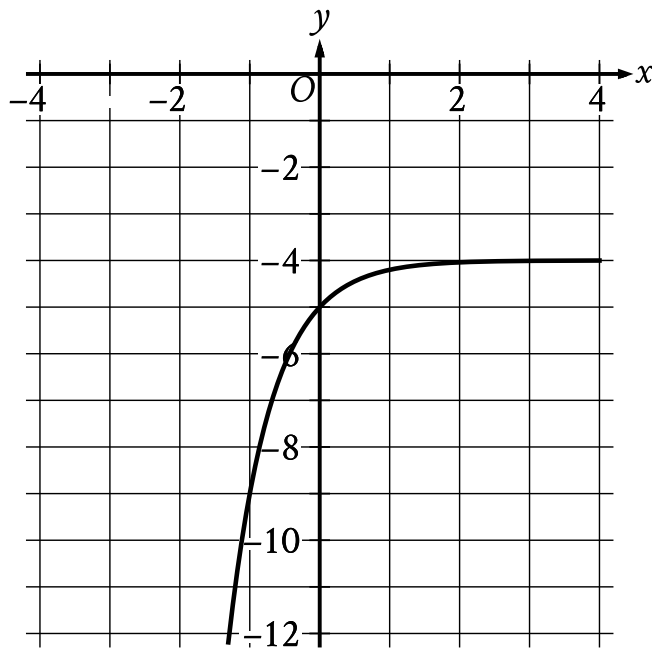
The graph of the polynomial function f , where $y = f(x)$, is shown. The y -intercept of the graph is 0, y . What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6

7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4.
 - In quadrant 3, the curve trends up sharply to point (0 comma negative 5).
 - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 4.
- The curve passes through the following points:
 - (negative 1 comma negative 9)
 - (0 comma negative 5)
 - (1 comma negative 4.2)

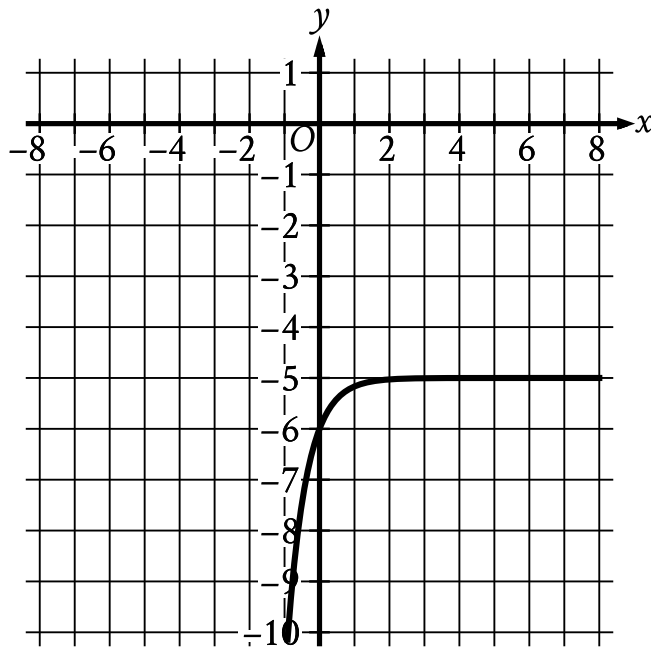
What is the y -intercept of the graph shown?

A. -1, -9

B. 0, -5

C. 0, -4

D. 0,0



- Moving from left to right:
 - The curve passes from quadrant 3 to quadrant 4.
 - In quadrant 3, the curve trends up sharply to point (0 comma negative 6).
 - In quadrant 4, the curve trends up gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
 - (0 comma negative 6)
 - (1 comma negative 5 and one sixth)

What is the y -intercept of the graph shown?

- A. 0, -6
- B. -6, 0
- C. 0, 0

D. -5, -5

The function $f(x) = 200,000(1.21)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online, where $0 < x \leq 10$. What is the best interpretation of the statement " $f(5)$ is approximately equal to 518,748" in this context?

- A. 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately 518,748 dollars.
- B. 5 years after the company started selling light bulbs online, its predicted annual revenue will have increased by a total of approximately 518,748 dollars.
- C. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5 times the predicted annual revenue for the previous year.
- D. When the company's predicted annual revenue is approximately 518,748 dollars, it is 5% greater than the predicted annual revenue for the previous year.

$$f(x) = 2(3^x)$$

For the function f defined above, what is the value of $f(2)$?

- A. 9
- B. 12
- C. 18
- D. 36